FLIGHT BOOKING SYSTEM USING DOUBLY LINKED LIST

```
#include <iostream>
#include <string>
#include <cstdlib>
using namespace std;

struct Booking
{
   int id;
   string name, source, destination;
   Booking *prev, *next;
};

class FlightBooking
{
   Booking *head;

public:
   FlightBooking()
   {
      head = NULL;
   }

   void bookFlight()
   {
      Booking *n = new Booking;

      cout << "Enter Booking ID: ";
      cin >> n->id;

      if (searchID(n->id))
```

```
    {
        cout << "Booking ID already exists!\n";
        delete n;
        return;
    }
    cout << "Passenger Name: ";
    cin >> n->name;
    cout << "Source: ";
    cin >> n->source;
    cout << "Destination: ";
    cin >> n->destination;

    n->prev = NULL;
    n->next = head;

    if (head != NULL)
        head->prev = n;

    head = n;

    cout << "Flight booked successfully!\n";
}

bool searchID(int id)
{
    Booking *t = head;

    while (t != NULL)
    {
        if (t->id == id)
            return true;
```

```
        t = t->next;
    }

    return false;
}

void display()
{
    Booking *t = head;

    if (t == NULL)
    {
        cout << "No bookings available.\n";
        return;
    }

    while (t != NULL)
    {
        cout << "\nID: " << t->id;
        cout << "\nName: " << t->name;
        cout << "\nSource: " << t->source;
        cout << "\nDestination: " << t->destination << "\n";

        t = t->next;
    }
}

void displayBackward()
{
    if (head == NULL)
    {
        cout << "No bookings available.\n";
```

```
        return;
    }

    Booking *t = head;

    while (t->next != NULL)
        t = t->next;

    cout << "\nBookings in reverse order:\n";

    while (t != NULL)
    {
        cout << t->id << " - " << t->name << "\n";
        t = t->prev;
    }
}

void search()
{
    int id;

    cout << "Enter Booking ID: ";
    cin >> id;

    Booking *t = head;

    while (t != NULL)
    {
        if (t->id == id)
        {
            cout << "Booking Found!\n";
            cout << "Passenger: " << t->name << "\n";
```

```
            cout << "Route: " << t->source
                 << " to " << t->destination << "\n";
            return;
        }

        t = t->next;
    }

    cout << "Booking not found.\n";
}

void cancel()
{
    int id;

    cout << "Enter Booking ID to cancel: ";
    cin >> id;

    Booking *t = head;

    while (t != NULL && t->id != id)
        t = t->next;

    if (t == NULL)
    {
        cout << "Booking not found.\n";
        return;
    }

    if (t->prev != NULL)
        t->prev->next = t->next;
    else
```

```
        head = t->next;

    if (t->next != NULL)
        t->next->prev = t->prev;

    delete t;

    cout << "Booking cancelled successfully!\n";
}

void memoryDemo()
{
    char *data = (char *)malloc(10 * sizeof(char));

    if (data == NULL)
    {
        cout << "Memory allocation failed!\n";
        return;
    }
    int *arr = (int *)calloc(5, sizeof(int));

    if (arr == NULL)
    {
        free(data);
        cout << "Memory allocation failed!\n";
        return;
    }

    cout << "\ncalloc array values: ";
    for (int i = 0; i < 5; i++)
        cout << arr[i] << " ";
```

```
        char *temp = (char *)realloc(data, 20 * sizeof(char));

        if (temp == NULL)
        {
            free(data);
            free(arr);
            cout << "\nReallocation failed!\n";
            return;
        }

        data = temp;

        cout << "\nMemory resized using realloc().\n";
        free(data);
        free(arr);

        cout << "Memory released using free().\n";
    }

    ~FlightBooking()
    {
        while (head != NULL)
        {
            Booking *t = head;
            head = head->next;
            delete t;
        }
    }
};

int main()
{
```

```
    FlightBooking f;
    int ch;

    do
    {
        cout << "\n--- FLIGHT BOOKING SYSTEM ---\n";
        cout << "1. Book Flight\n";
        cout << "2. Display Bookings\n";
        cout << "3. Search Booking\n";
        cout << "4. Cancel Booking\n";
        cout << "5. Display Backward\n";
        cout << "6. Demonstrate malloc/calloc/realloc/free\n";
        cout << "7. Exit\n";
        cout << "Enter choice: ";
        cin >> ch;

        switch (ch)
        {
            case 1: f.bookFlight(); break;
            case 2: f.display(); break;
            case 3: f.search(); break;
            case 4: f.cancel(); break;
            case 5: f.displayBackward(); break;
            case 6: f.memoryDemo(); break;
            case 7: cout << "Thank you!\n"; break;
            default: cout << "Invalid choice!\n";
        }

    } while (ch != 7);

    return 0;
}
```